12th HACKERS KA JUGAAD

Ray Optics and Optical Instruments

Reflection of Light

i) angle of incidence = angle of reflection
$\angle i = \angle r$

ii) The incident ray, the reflected ray and the normal at point of incidence all lie in the same plane.

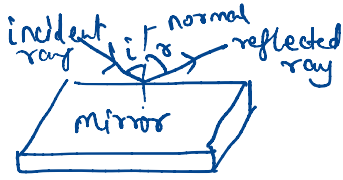
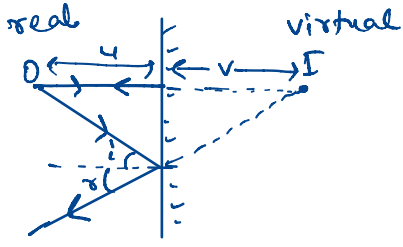
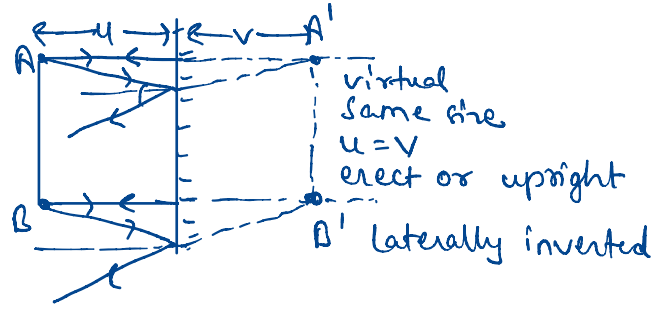
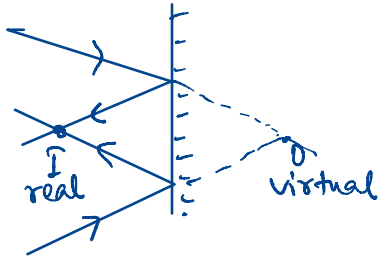


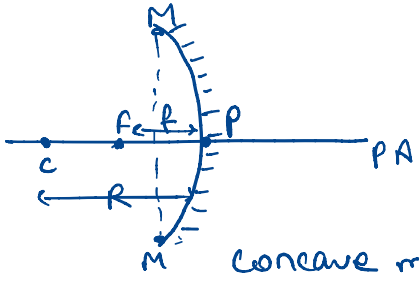
Image formed by a plane Mirror







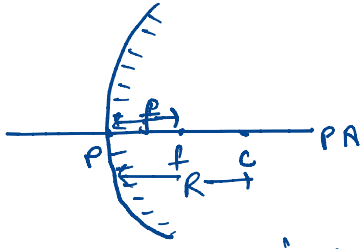
Spherical Mirrors



Concave mirror

$f < 0$

$R < 0$



Convex mirror

$f > 0$

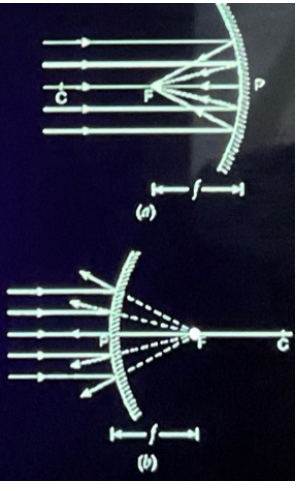
$R > 0$.

Few Definitions

Principal focus. The point at which, a narrow beam of light incident on the mirror parallel to its principal axis, after reflection from the mirror, meets or appears to come from, is called the principal focus of the mirror. It is represented by the point F.

Focal length. The distance between the pole and the principal focus of the mirror is called the focal length of the mirror.

Focal plane. The vertical plane passing through the principal focus and perpendicular to the principal axis is called focal plane.



Sign Convention

i) Object $\underset{-ve}{\overset{+ve}{\rightleftarrows}}$ Mirror / lens

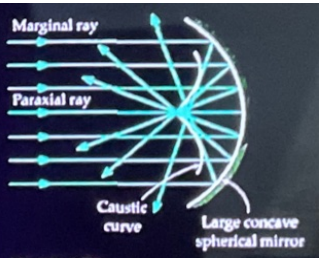
ii) $\uparrow +ve$ — PA — $\downarrow -ve$

iii) Measured from pole.

Spherical Aberration

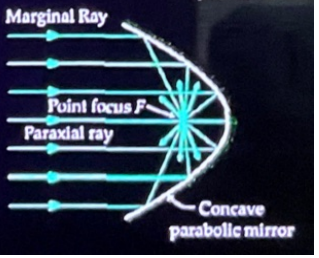
The inability of a spherical mirror of large aperture to bring all the rays of wide beam of light falling on it to focus at a single point is called spherical aberration.

- Only the paraxial rays are focused at the principal focus F.
- The marginal rays meet the principal axis at a point closer to the pole than the principal focus.



Spherical aberration can be reduced by following methods :

1. By using spherical mirrors of small apertures.
2. By using stoppers so as to cut off the marginal rays.
3. By using parabolic mirrors.



Relation b/w f and R

$\angle ABC = \angle BCF$ (Alternate angles)

$i = \alpha$ —①

But $i = r$ (law of reflection) —②

from eq^n ① & ②

$i = r = \alpha$

In $\triangle BFC$

$\alpha = r$

$FB = CF$ —③

Because mirror is of small aperture. B & P are very close

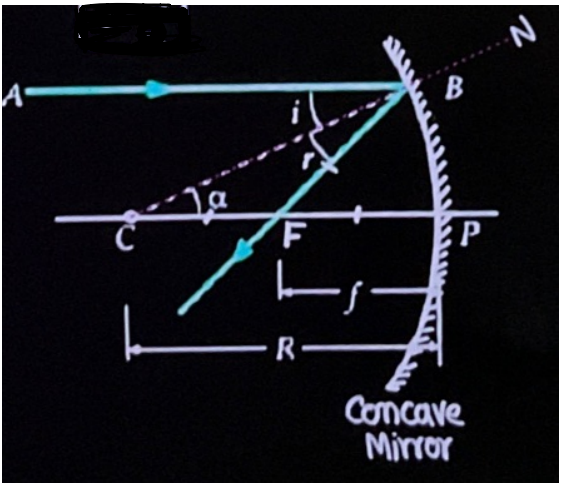
$FB \approx FP$ —④

from eq. ② & ④

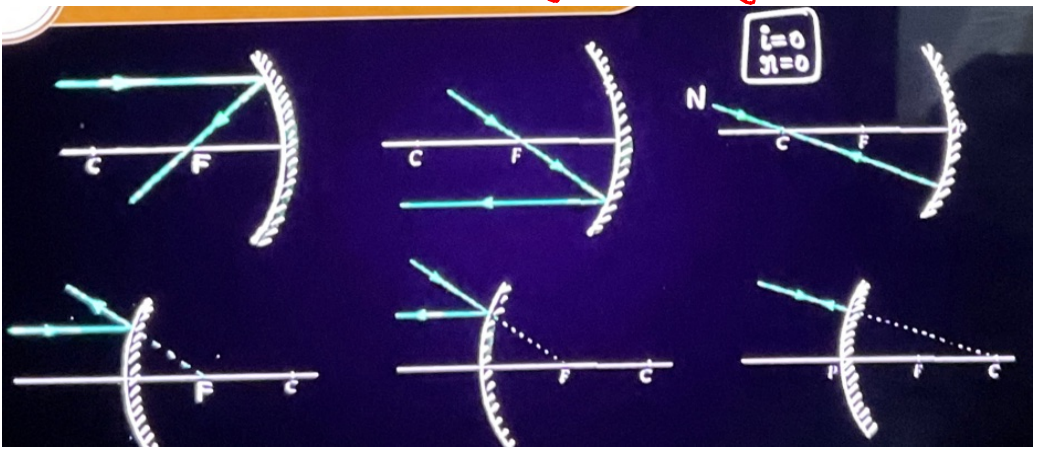
$CF = FP$

Hence F is the mid point of CP.

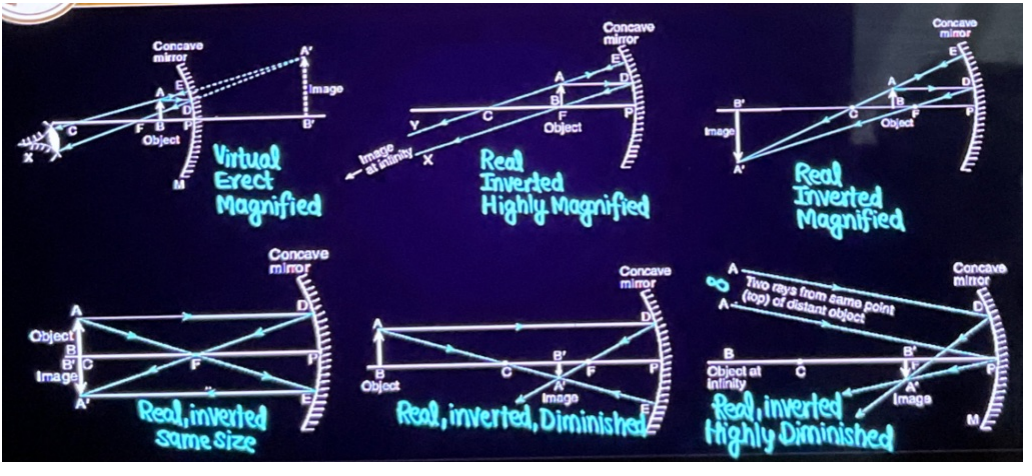
$f = R/2$.



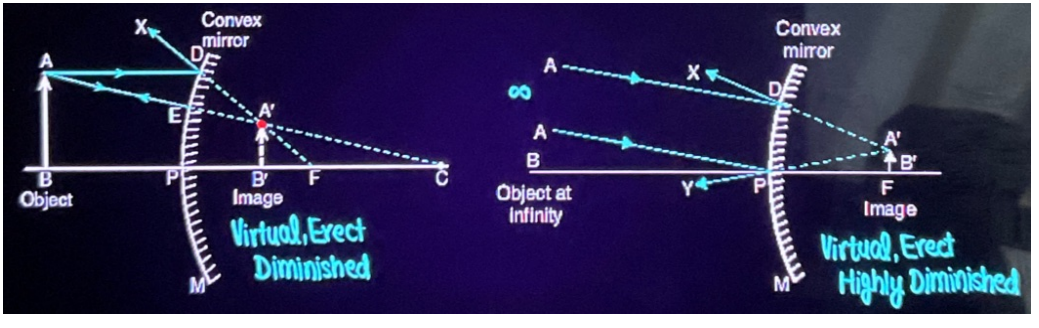
Rules for Drawing Images



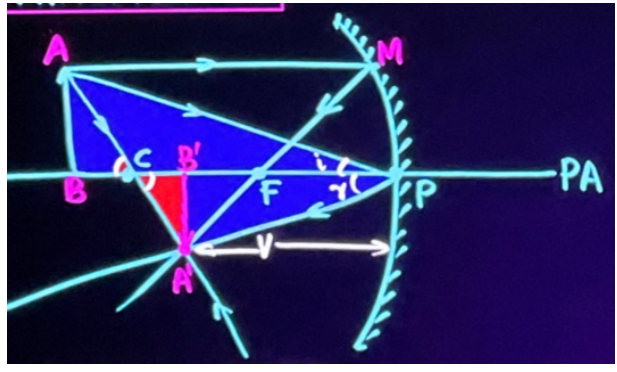
Formation of images by Concave Mirrors



Formation of image by convex mirror



Mirror Formula



In $\triangle ABC$ & $\triangle A'B'C$

$\angle B = \angle B' = 90^\circ$

$\angle ACB = \angle A'CB'$

$\triangle ABC \sim \triangle A'B'C$

$$\frac{A'B'}{AB} = \frac{B'C}{BC} = \frac{CP - B'P}{BP - CP} = \frac{-R-(-v)}{-u-(-R)} = \frac{-R+v}{-u+R} \quad -(1)$$

In $\triangle ABP$ & $\triangle A'B'P$

$\angle B = \angle B' = 90^\circ$

$\angle APB = \angle A'PB'$

$\triangle ABP \sim \triangle A'B'P$

$$\frac{A'B'}{AB} = \frac{B'P}{BP} = \frac{-v}{-u} = \frac{v}{u} \quad -(2)$$

from eq^n (1) & (2)

$$\frac{-R+v}{-u+R} = \frac{v}{u}$$

$$-uR + uv = -uv + Rv$$

$$2uv = Rv + uR$$

$$2uv = R(v+u)$$

$$\frac{2}{R} = \frac{v+u}{uv}$$

$$\frac{2}{R} = \frac{1}{u} + \frac{1}{v}$$

$$f = \frac{R}{2}$$

$$R = 2f$$

$$\frac{2}{2f} = \frac{1}{v} + \frac{1}{u}$$

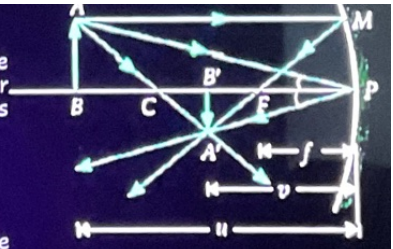
$$\boxed{\frac{1}{f} = \frac{1}{v} + \frac{1}{u}}$$

Magnification

$$m = \frac{h_i}{h_o} = \frac{A'B'}{AB} = \frac{-v}{u}$$

Linear magnification The ratio of the height of the image to that of the object is called linear or transverse magnification or just magnification and is denoted by m.

$$m = \frac{\text{Height of image}}{\text{Height of object}} = \frac{h_2}{h_1}$$



Concave mirror Fig. shows the ray diagram for the formation of image $A'B'$ of a finite object AB by a concave mirror.

Now, $\Delta APB \sim \Delta A'PB'$

$$\therefore \frac{A'B}{AB} = \frac{BP}{BP}$$

$$\frac{-h_2}{h_1} = \frac{-v}{-u}$$

Magnification,

$$m = \frac{h_2}{h_1} = -\frac{v}{u}$$

Refraction of light

the phenomenon of change in the path of light as it passes obliquely from one transparent medium to another is called refraction of light.

Medium

- rarer
  - $v \uparrow$
  - vacuum → R
  - Air → D
  - Air → R
  - water → D
- denser
  - $v \downarrow$
  - water → R
  - Glass → D
  - Glass → R
  - Diamond → D

Refractive Index

$$\mu \text{ or } n = \frac{c}{v} = \frac{3 \times 10^8}{v}$$

$$n = \frac{1}{v}$$

$$n_{denser} > n_{rarer}.$$

for any wave,

$$v = f\lambda$$

↳ property of source.

frequency remains same during reflection and refraction.

$$n = \frac{c}{v} = \frac{f\lambda_v}{f\lambda_m} = \frac{\lambda_v}{\lambda_m}$$

$$\lambda_m = \frac{\lambda_v}{n}$$

Vacuum	Medium
λ_m	λ_m
400 nm	200 nm

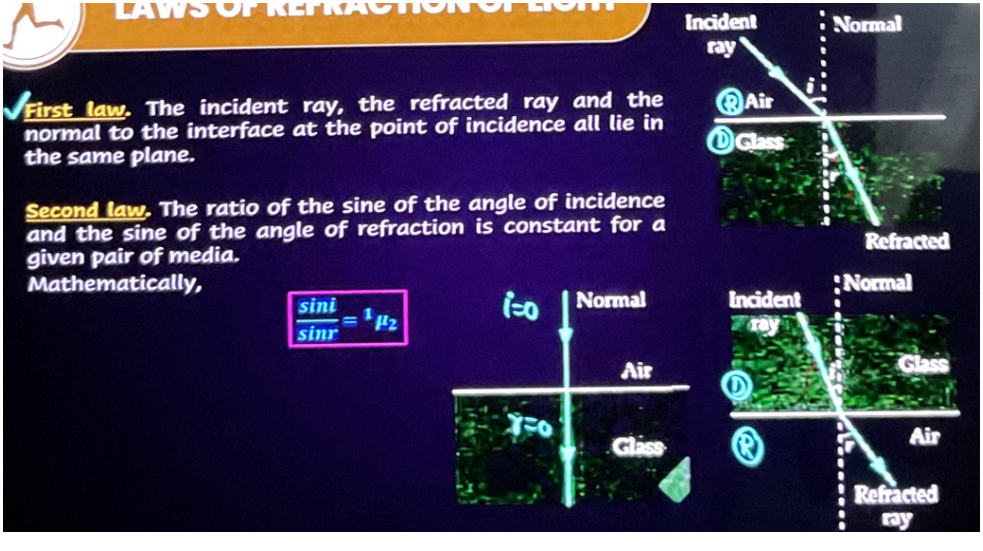
$$n = \frac{400}{200} = 2$$

air → water

$n = 1$	$n = 4/3$
400 nm	$\lambda_w = ?$

$$\lambda_w = \frac{\lambda_a}{n_w} = \frac{400}{4/3} = 300 \text{ nm}$$

Laws of Refraction



Refractive Index

The refractive index of a medium for a light of given wavelength may be defined as the ratio of the speed of light in vacuum to its speed in that medium.

$$\text{Refractive index} = \frac{\text{Speed of light in vacuum}}{\text{Speed of light in medium}}$$

$$\mu = \frac{c}{v}$$

Refractive index of a medium with respect to vacuum is also called absolute refractive index.

factors on which the refractive index of a medium depends

1. Nature of the medium.
2. Wavelength of the light used. ($n \propto \frac{1}{\lambda}$)
3. Temperature. T↑ Rarer↑
4. Nature of the surrounding medium.

Refraction through a rectangular glass slab



In $\triangle QUR$,

$$\sin(i-r) = \frac{d}{QR} \quad -(1)$$

In $\triangle QTR$,

$$\cos r = \frac{t}{QR} \quad -(2)$$

eq-(1) & (2)

$$\frac{\sin(i-r)}{\cos r} = \frac{d}{t}$$

$$d = t\,\frac{\sin(i-r)}{\cos r} \quad -(3)$$

By snell's law,

$$\frac{\sin i}{\sin r} = \frac{n_g}{n_a} = \frac{n_g}{1}$$

$$\sin r = \frac{\sin i}{n_g} \quad -(4)$$

$$\cos r = \sqrt{1-\sin^2 r}$$

$$\cos r = \sqrt{1-\left(\frac{\sin i}{n_g}\right)^2} \quad -(5)$$

$$\sin(A-B) = \sin A \cos B - \cos A \cdot \sin B$$

from eq. (3)

$$d = \frac{t}{\cos r}(\sin i \cos r - \cos i \sin r)$$

$$d = t\left(\sin i - \cos i \frac{\sin r}{\cos r}\right)$$

$$d = t\left(\sin i - \cos i \frac{\sin i/n_g}{\sqrt{1-\frac{\sin^2 i}{n_g}}}\right)$$

$$\boxed{d = t\left[\sin i - \frac{\cos i \sin i}{\sqrt{n_g^2 - \sin^2 i}}\right]}$$

$$\boxed{\begin{array}{l} d \propto t \\ d \propto \frac{1}{n_g} \end{array}}$$

Normal Shift

In $\triangle OAB$,

$$\sin i = \frac{AB}{OB} \quad -(1)$$

In $\triangle IAB$,

$$\sin r = \frac{AB}{IB} \quad -(2)$$

By Snell's law,

$$\frac{\sin i}{\sin r} = \frac{n_r}{n_d}$$

$$\frac{AB/OB}{AB/IB} = \frac{n_m}{n_d}$$

$$\frac{IB}{OB} = \frac{n_r}{n_d} \quad -(3)$$

A & B are very close to each other

$OB \simeq OA$

$IB \simeq IA$

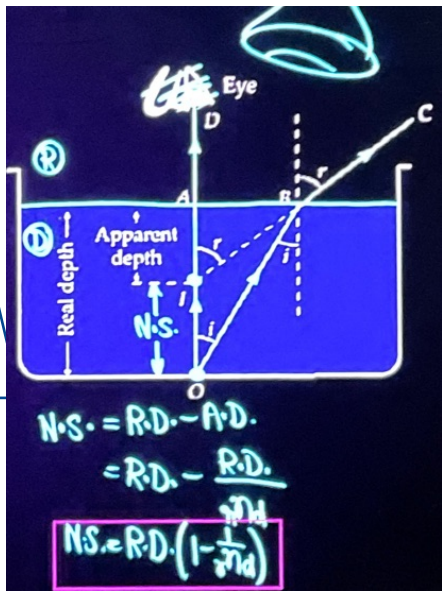
from eq (3)

$$\frac{IA}{OA} = \frac{n_r}{n_d}$$

$$\frac{AD}{RD} = \frac{n_m}{n_d}$$

$$A.D = \frac{RD}{n_d/n_m}$$

$$\boxed{A.D = \frac{R.D.}{n_d}}$$



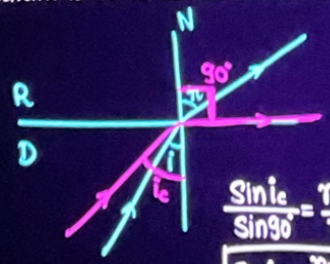
$$N.S. = R.D. - A.D.$$

$$= R.D. - \frac{R.D.}{n_d}$$

$$\boxed{N.S. = R.D.\left(1 - \frac{1}{n_d}\right)}$$

Critical Angle

The angle of incidence in the denser medium for which the angle of refraction in the rarer medium is 90° is called critical angle of the denser medium and is denoted by i_c.



$$\frac{\sin i_c}{\sin 90^\circ} = \frac{n_r}{n_d}$$

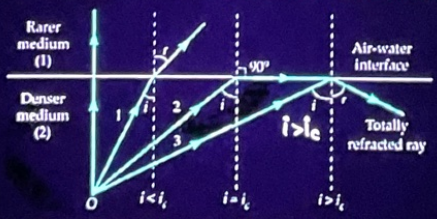
$$\sin i_c = \frac{n_r}{n_d}$$

$$\sin i_c = \frac{1}{\frac{n_d}{n_r}}$$

$$\sin i_c = \frac{1}{n_d}$$

Total Internal Reflection

The phenomenon in which a ray of light travelling at an angle of incidence greater than the critical angle from denser to a rarer medium is totally reflected back into the denser medium is called total internal reflection.



Necessary conditions for total internal reflection:

- Light must travel from an optically denser to an optically rarer medium.
- The angle of incidence in the denser medium must be greater than the critical angle for the two media.

Application of total internal reflection

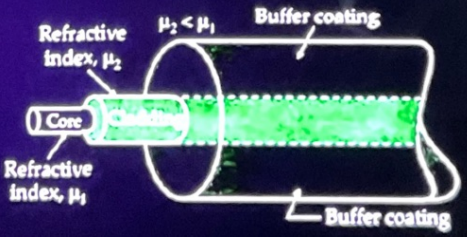
SPARKLING OF DIAMOND : The brilliancy of diamonds is due to total internal reflection. As the refractive index of diamond is very large, its critical angle is very small, about 24.4°. The faces of diamond are so cut that the light entering the crystal suffers total internal reflections repeatedly, and hence gets collected inside but it comes out through only a few faces. Hence the diamond sparkles when seen in the direction of emerging light.

MIRAGE : It is an optical illusion observed in deserts or over hot extended surfaces like a coal-tarred road, due to which a traveler sees a shimmering pond of water some distance ahead of him and in which the surrounding objects like trees, etc, appear inverted.



OPTICAL FIBRES : The working of optical fibres is based on the phenomenon of total internal reflection. An optical fibre is a hair-thin long strand of quality glass or quartz surrounded by a glass coating of slightly lower reflective index. It is used as a guided medium for transmitting an optical signal from one place to another.

$n_{core} > n_{cladding}$

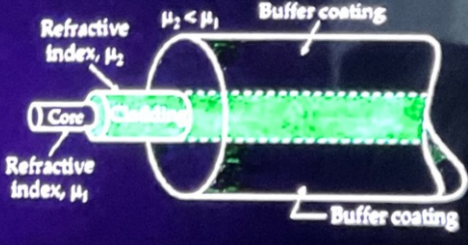


Construction

(i) **Core.** The central cylindrical core is made of high quality glass/silica/plastic of refractive index μ_1 and has a diameter about 10 to 100μm.

(ii) **Cladding.** The core is surrounded by a glass / plastic jacket of refractive index $\mu_2 < \mu_1$. In a typical optical fibre, the refractive indices of core and cladding may be 1.52 and 1.48 respectively.

(iii) **Buffer coating.** For providing safety and strength, the core cladding of optical fibres is enclosed in a plastic jacket.



1. **As a light pipe,** optical fibres are used in medical and optical examination. A light pipe is inserted into the stomach through the mouth. Light transmitted through the outer layers of the light pipe is scattered by the various parts of stomach into the central portion of the light pipe to produce a final image with excellent details. The technique is called endoscopy.

2. They are used in **transmitting and receiving electrical signals in telecommunication.** The electrical signals are first converted to light by suitable transducers. Each fibre can transmit about 2000 telephone conversations without much loss of intensity.

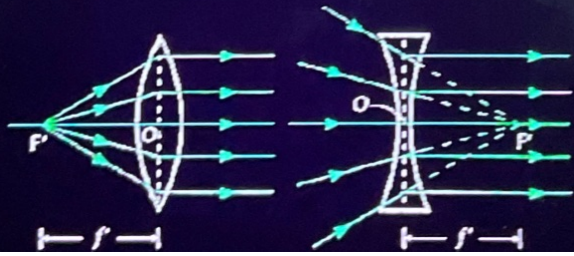
3. They are **used for transmitting optical signals and two dimensional pictures.**

4. In the form of **photometric sensors,** they are used for measuring the blood flow in the heart.

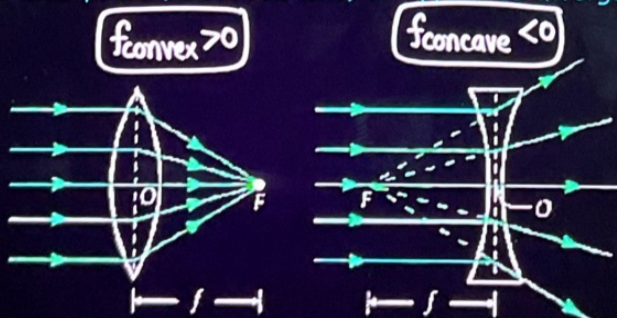
5. In the form of **refractometers,** they are used to measure refractive indices of liquids.

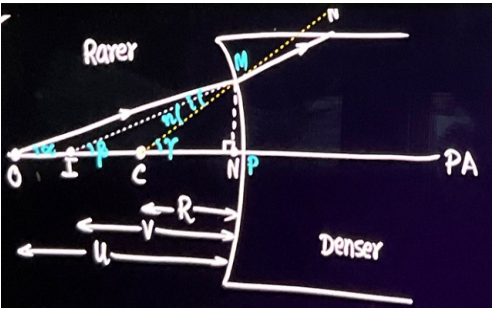
Definitions in connection with spherical lens

FIRST PRINCIPAL FOCUS : It is a fixed point on the principal axis such that rays starting from this point (in convex lens) or appearing to go towards this point (in concave lens), after refraction through the lens, become parallel to the principal axis. It is represented by F_1 or F'.



SECOND PRINCIPAL FOCUS : It is a fixed point on the principal axis such that the light rays incident parallel to the principal axis, after refraction through the lens, either converge to this point (in convex lens) or appear to diverge from this point (in concave lens).





In $\triangle MCO$, $\gamma = \alpha + i \Rightarrow i = \gamma - \alpha$ — (1)

In $\triangle MCI$, $\gamma = \beta + r \Rightarrow r = \gamma - \beta$ — (2)

By Snell's law,

$$\frac{\sin i}{\sin r} = \frac{n_d}{n_r}$$

$$n_r \sin i = n_d \sin r \quad \text{— (3)}$$

Considering $\alpha, \beta, \gamma, i, r$ very small

$\sin i = i$

$\sin r = r$

from eq. (3)

$n_r i = n_d r$

from (1) & (2)

$n_r(\gamma - \alpha) = n_d(\gamma - \beta)$

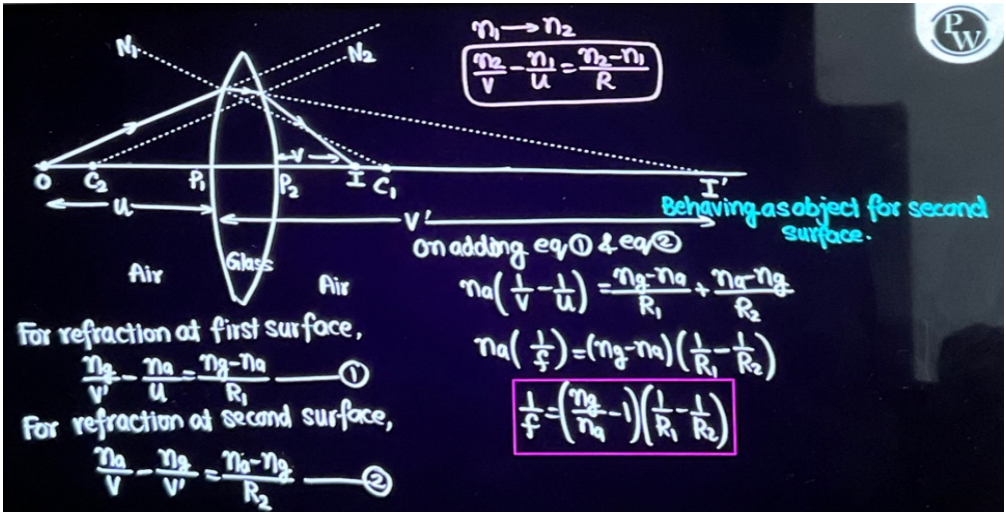
$n_r(\tan\gamma - \tan\alpha) = n_d(\tan\gamma - \tan\beta)$

$$n_r\left(\frac{MN}{-R} - \frac{MN}{-u}\right) = n_d\left(\frac{MN}{-R} - \frac{MN}{-v}\right)$$

$$\frac{n_r}{-R} + \frac{n_r}{u} = \frac{n_d}{-R} + \frac{n_d}{v}$$

$$\frac{n_d - n_r}{R} = \frac{n_d}{v} - \frac{n_r}{u}$$

Lens Maker's formula



Liquid

$$\frac{1}{f} = \left(\frac{n_g}{n_l} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

case - I $n_g > n_l$

nature of lens will not change

case-II $n_g < n_l$

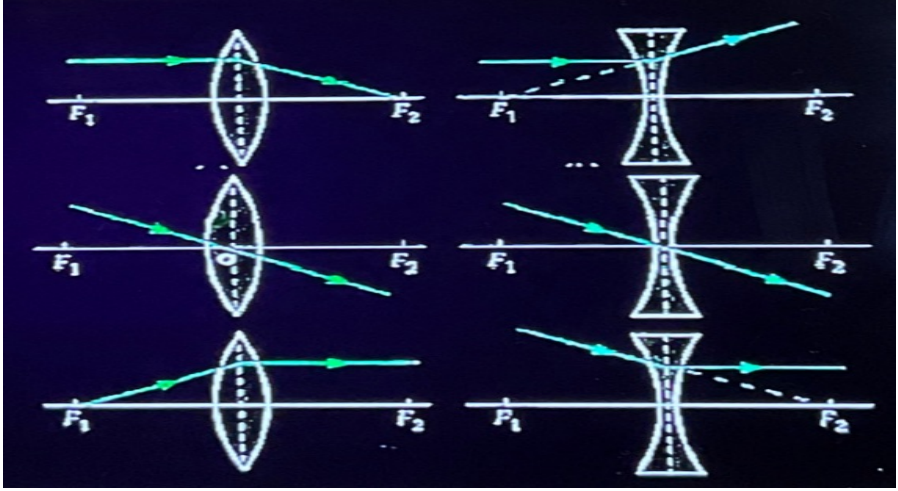
nature of lens will change

Case III $n_g = n_l$

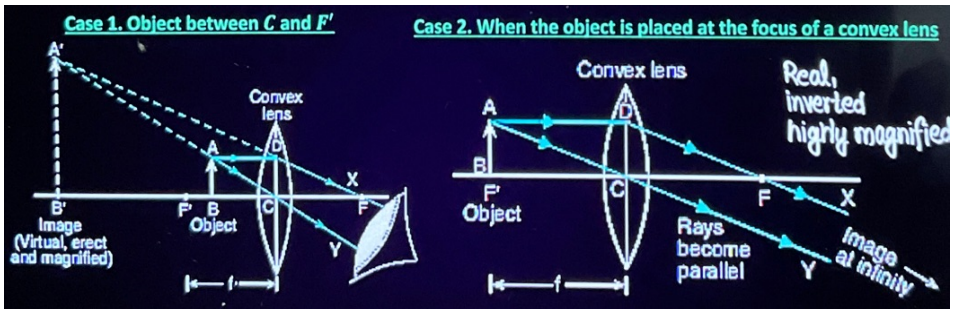
$f = \infty$

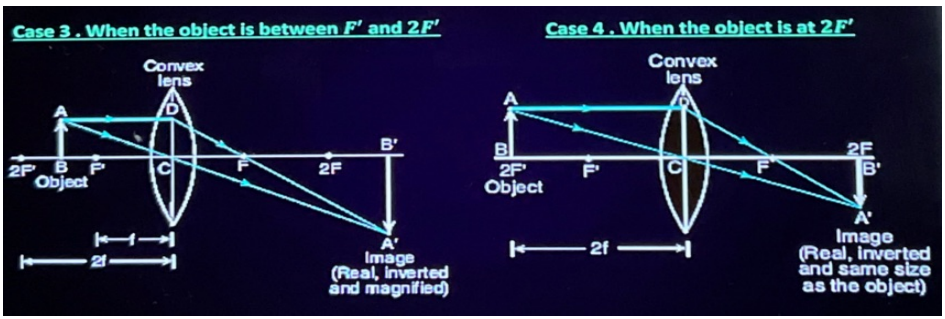
lens will behave as transparent glass sheet (Disappear).

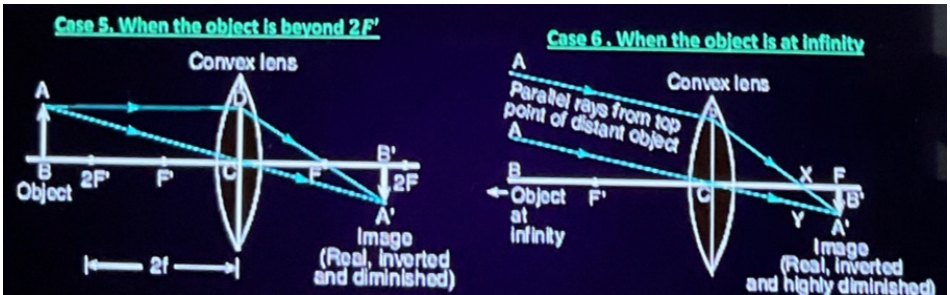
Rules for images formed by spherical lenses



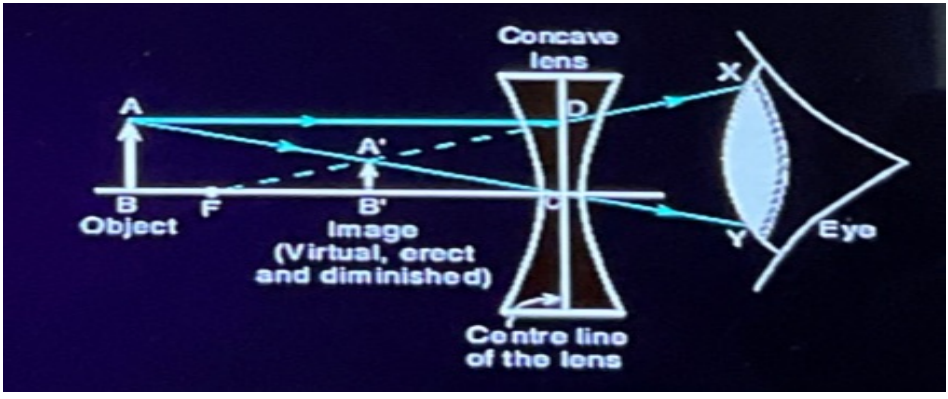
formation of images by a convex lens







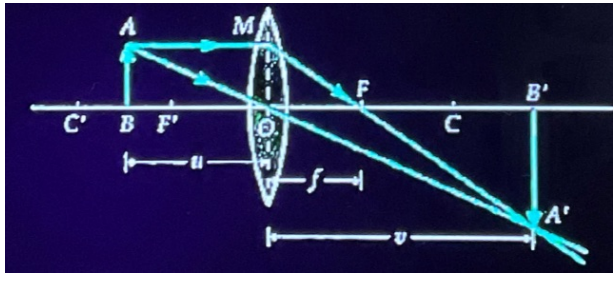
formation of Images by concave lens



Thin lens formula.

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

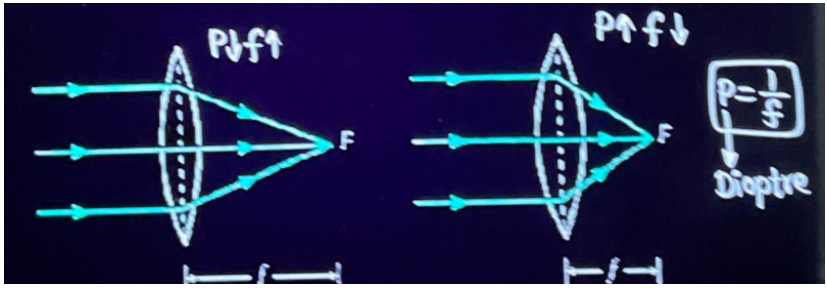
$$m = \frac{h_i}{h_o} = \frac{v}{u}$$



Power of a lens

→ Measure of the degree of convergence or divergence of the light rays falling on it.

→ Smaller the focal length, more power.



Combination of thin lenses

Using thin lens formula, $\frac{1}{f_1}=\frac{1}{v}-\frac{1}{u}$ —①

The image I' acts as a virtual object ($u=v'$) for the second lens L_2 which finally forms its real image I at distance v. Thus $\frac{1}{f_2}=\frac{1}{v}-\frac{1}{v'}$ —②

Adding equations (1) and (2), we get

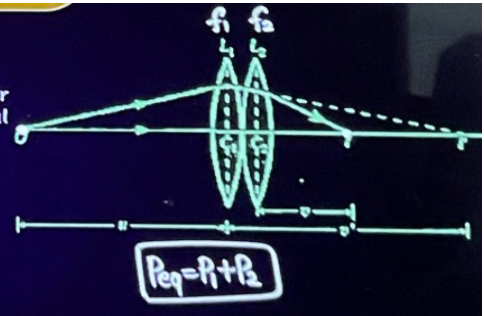
$$\frac{1}{f_1}+\frac{1}{f_2}=\frac{1}{v}-\frac{1}{u}$$

if f is the equivalent focal length, then $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$

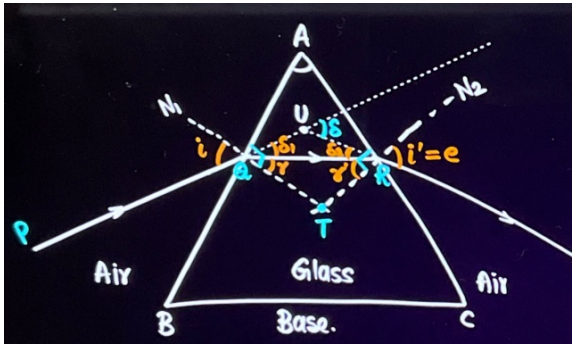
From equations, we find that

$$\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}$$

∴ Equivalent power, $P=P_1+P_2$



Refraction through a prism



In Δ QUR,

$\delta=\delta_1+\delta_2$

$\delta=(i-r)+(i'-r')$

$\delta=(i+i')-(r+r')$ —①

In Δ QTR,

$r+r'+\angle T=180°$ —②

In □ AQTR

$\angle A+90°+\angle T+90°=360°$

$\angle A+\angle T=180°$ —③

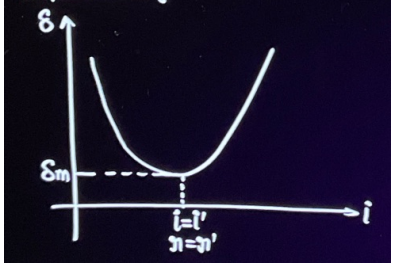
from eq. ② & ③

$r+r'=\angle A$ —④

from eq. ① & ④

$\delta=(i+i')-\angle A$

$\delta+\angle A=i+i'$ —⑤



experimentally,

for minm deviation

from eq. (1)

$2r = \angle A$

$r = \frac{\angle A}{2}$ — (6)

from eq. (1)

$\delta = 2i - \angle A$

$\frac{\angle A + \delta_m}{2} = i$

By snell's law

$\frac{\sin i}{\sin r} = \frac{n_g}{n_a}$

$$\frac{\sin\left(\frac{\angle A + \delta_m}{2}\right)}{\sin\left(\angle A/2\right)} = a n_g$$

Deviation through a prism of small angle

If $\angle A$ is very small,

$$\frac{\sin \frac{\angle A + \delta_m}{2}}{\sin \frac{\angle A}{2}} = a n_g$$

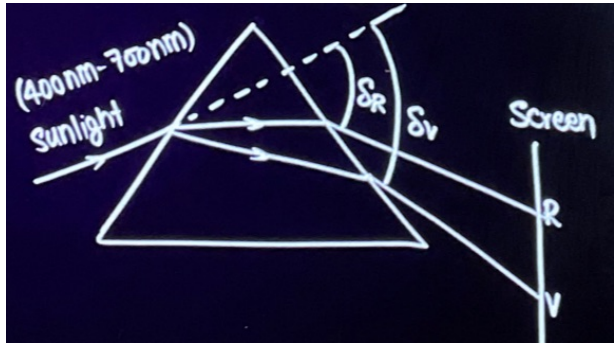
$$\frac{\frac{\angle A + \delta_m}{2}}{\angle A/2} = a n_g$$

$$1 + \frac{\delta_m}{\angle A} = a n_g$$

$$\delta_m = (a n_g - 1)\angle A$$

$\delta_m = (n-1) A$

$\delta \propto n \quad \alpha \frac{1}{\lambda} \quad \alpha \frac{1}{\lambda}$



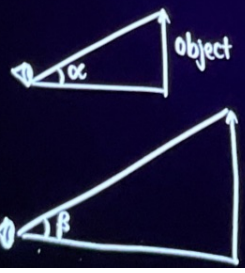
Magnifying Power

Magnifying power of a simple microscope is defined as the ratio of the angles subtended by the image and the object at the eye, when both are at the least distance of distinct vision from the eye.

$$\text{Magnifying Power} = \frac{\text{Angle subtended by image at least distance of distinct vision}}{\text{Angle subtended by object at least distance of distinct vision}}$$

$M = \frac{\beta}{\alpha}$ — Angular magnification

$m = \frac{h_i}{h_o}$ — linear magnification



Simple Microscope

A simple microscope or a magnifying glass is just a convex lens of short focal length, held close to the eye.

Case I

$$M = \frac{\beta}{\alpha} = \frac{\tan\beta}{\tan\alpha} = \frac{AB/BO}{AB/D} = \frac{-D}{-u} = \frac{D}{u}$$

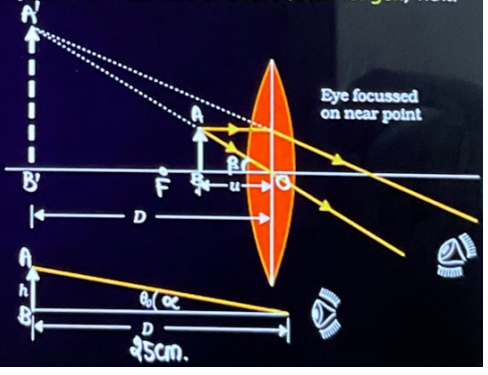
By lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{-D} - \frac{1}{-u}$$

$$\frac{1}{f} = -\frac{1}{D} + \frac{1}{u}$$

$$\frac{D}{f} = -1 + \frac{D}{u} \Rightarrow \frac{D}{u} = 1 + \frac{D}{f} \Rightarrow \boxed{M = 1 + \frac{D}{f}}$$



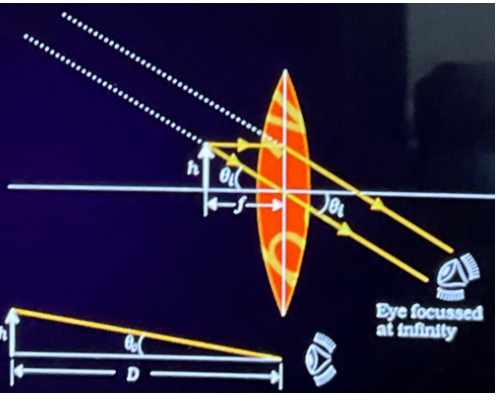
Case II

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

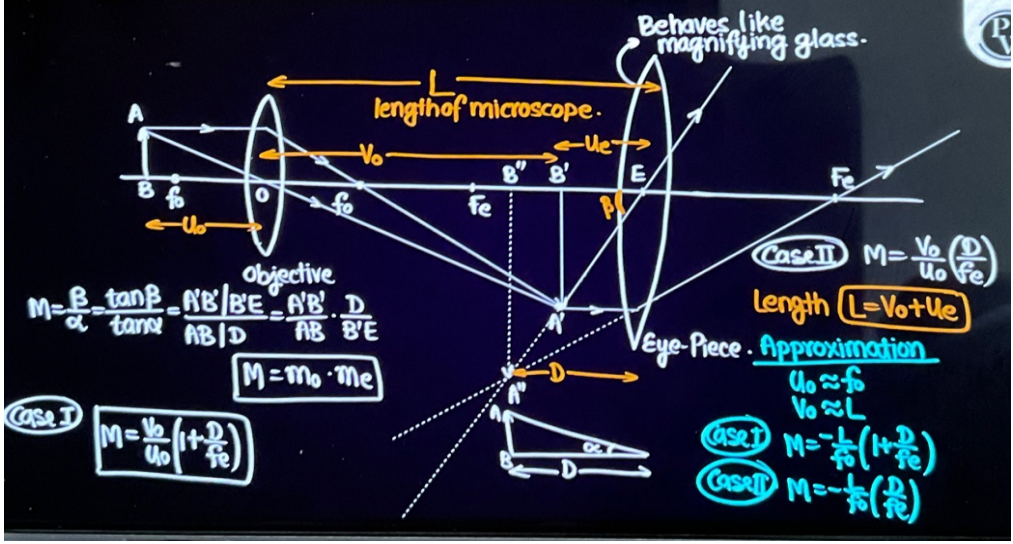
$$\frac{1}{f} = \frac{1}{\infty} - \frac{1}{-u}$$

$$\frac{D}{f} = \frac{D}{u}$$

$$\boxed{M = \frac{D}{f}}$$



Compound Microscope



Astronomical Telescope

A telescope is an optical device which enables us to see distant objects clearly. It provides an angular magnification of the distant objects.

DIFFERENT TYPES OF TELESCOPE

- **Refracting telescopes**. These make use of lenses to view distant objects.
  - (a) **Astronomical telescope**. It is used to see heavenly objects like the sun, stars, planets, etc. The final image formed is inverted one which is immaterial in the case of heavenly bodies because of their round shape.
  - (b) **Terrestrial telescope**. It is used to see distant objects on the surface of the earth. The final image formed is erect one. This is an essential condition of viewing the objects on earth's surface correctly.
- **Reflecting telescopes**. These make use of converging mirrors to view the distant objects. For example, **Newtonian** and **Cassegrain telescopes**.

∴ Magnifying power,

$$m = \frac{\beta}{\alpha} = \frac{\tan\beta}{\tan\alpha} = \frac{A'B'/B'E}{A'B'/OB'} = \frac{OB'}{B'E}.$$

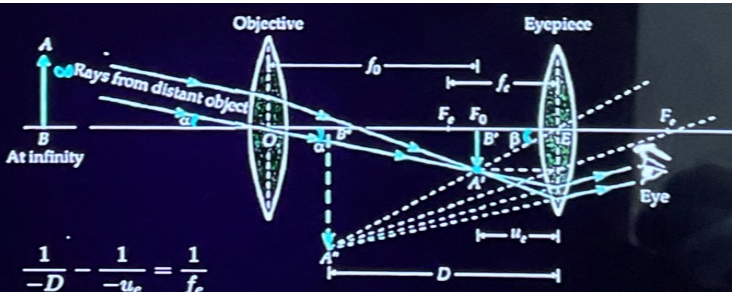
$$\therefore \quad \boxed{m = -\frac{f_0}{u_e}}$$

Again, for the eyepiece :

$\boxed{u = -u_e}$ and $\boxed{v = -D}$

As

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$



$$\frac{1}{-D} - \frac{1}{-u_e} = \frac{1}{f_e}$$

$$\frac{1}{u_e} = \frac{1}{f_e} + \frac{1}{D} = \frac{1}{f_e}\left(1 + \frac{f_e}{D}\right)$$

$$\boxed{m = -\frac{f_0}{f_e}\left(1 + \frac{f_e}{D}\right)}$$

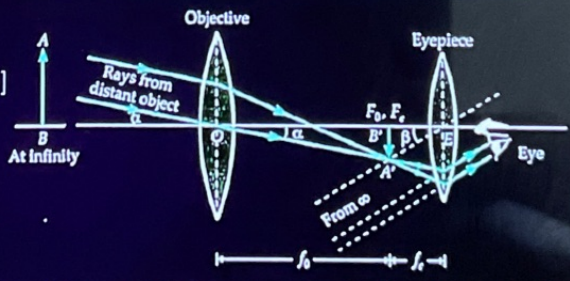
ASTRONOMICAL TELESCOPE IN NORMAL ADJUSTMENT

$\therefore$ Magnifying power,

$$m = \frac{\beta}{\alpha} = \frac{\tan\beta}{\tan\alpha} \quad [\because \alpha, \beta \text{ are small angles}]$$

$$= \frac{A'B'/B'E}{A'B'/OB'} = \frac{OB'}{B'E}$$

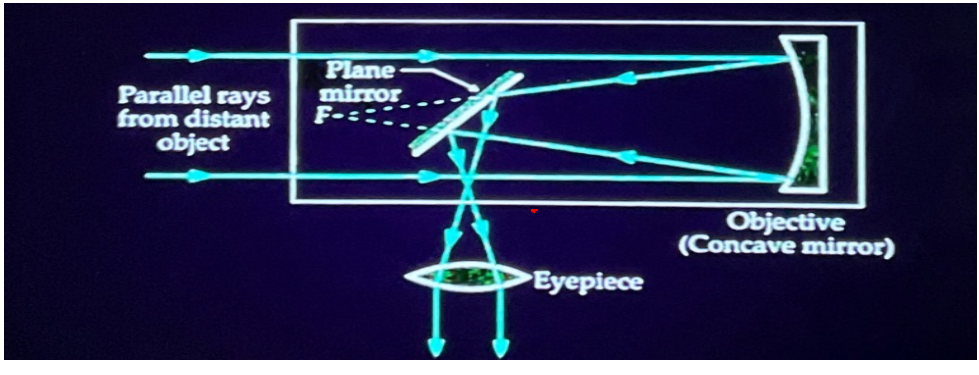
$$M = \frac{f_o}{-f_e}$$



Applying new Cartesian sign convention, $OB' = +f_o$ = Distance of $A'B'$ from the objective along the incident light
$B'E = -f_e$ = Distance of $A'B'$ from the eyepiece against the incident light

$$\therefore \quad m = -\frac{f_o}{f_e}$$

Newtonian Reflecting Telescope



Cassegrain reflecting Telescope

It consists of a large concave paraboloidal (primary) mirror having a hole at its centre. There is a small convex (secondary) mirror near the focus of the primary mirror. The eyepiece is placed on the axis of the telescope near the hole of the primary mirror.

